

DS555 - Applied Multivariate Statistics (Mini Project)

Hair, Anderson, and Tatham
Company - (HATCO) Customer
Satisfaction Survey and Analysis



(HATCO) Customer Satisfaction Analysis

Team Members:

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(HATCO) Customer Satisfaction Analysis

Team Contribution:

- Section (1): Introduction and Variables
(Mahmoud Abdelaziz)
- Section (2): Data Visualization and Descriptive Measures
(Taha Mohamed Taha)
- Section (3): Assumption validation
(Normality, Homogeneity)
(Mahmoud Abdelaziz, Eman Masry)
- Section (3): ANOVA Model
(Ahmed Ezzat Mostafa, Eman Masry)
- Section (3): MANOVA
(Eslam Kamel Morsi, Taha Mohamed Taha)
- Python Coding
(Ahmed Ezzat Mostafa)
- Documenting & Presentation
(Eslam Kamel Morsi, Taha Mohamed Taha)

(HATCO) Customer Satisfaction Analysis



DS555 - Applied Multivariate Statistics (Mini Project)

Section (1):
Introduction and Variables

Section (2):
Data Visualization and Descriptive Measures

Section (3):
Modeling and in-depth Analysis
One-Way ANOVA & One-Way MANOVA

Section (1): Introduction and Variables

Section (1): Introduction and Variables



HATCO Dataset Documentation:

Multivariate Data Analysis, Seventh edition

HATCO has been utilized in past versions of the textbook and provides a simplified set of variables amenable to all of the basic multivariate techniques. It was obtained from the Hair, Anderson, and Tatham Company (HATCO), a large (though nonexistent) industrial supplier. It represents surveys of HATCO customers that were collected through an established marketing research firm.

Section (1): Introduction and Variables

HATCO Customer Satisfaction Study

	Variable Name	Role in Study	Variable Type	Level of Measurement
	X1 Delivery speed	Original Variable	Quantitative	ratio
	X2 Price level	Potential (DV)	Quantitative	ratio
	X3 Price flexibility	Original Variable	Quantitative	ratio
	X4 Manufacturer's image	Potential (DV)	Quantitative	ratio
	X5 Service level	Original Variable	Quantitative	ratio
	X6 Salesforce's image	Potential (DV)	Quantitative	ratio
	X7 Product quality	Original Variable	Quantitative	ratio

Section (1): Introduction and Variables

HATCO Customer Satisfaction Study

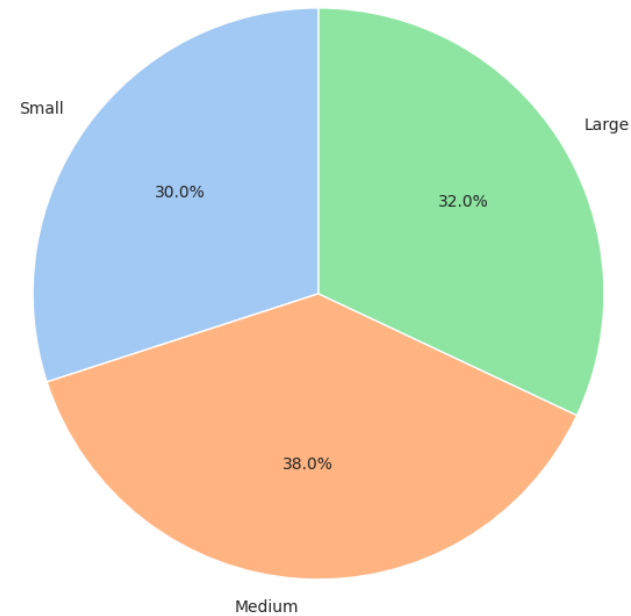
	Variable Name	Role in Study	Variable Type	Level of Measurement
	X8 Size of firm	Potential (IV)	Qualitative (3 Groups)	ordinal
	X9 Usage level	Original Variable	Quantitative	ratio
	X10 Satisfaction level	Potential (DV)	Quantitative	ratio
	X11 Specification buying	Original Variable	Qualitative (2 Groups)	nominal
	X12 Structure of procurement	Original Variable	Qualitative (2 Groups)	nominal
	X13 Type of industry	Original Variable	Qualitative (2 Groups)	nominal
	X14 Type of buying situation	Potential (IV)	Qualitative (3 Groups)	nominal

Section (1): Introduction and Variables

Firm size:

firm_size	Count	Percent %
small	30	30
medium	38	38
large	32	32

Figure 1: Firm Size Distribution

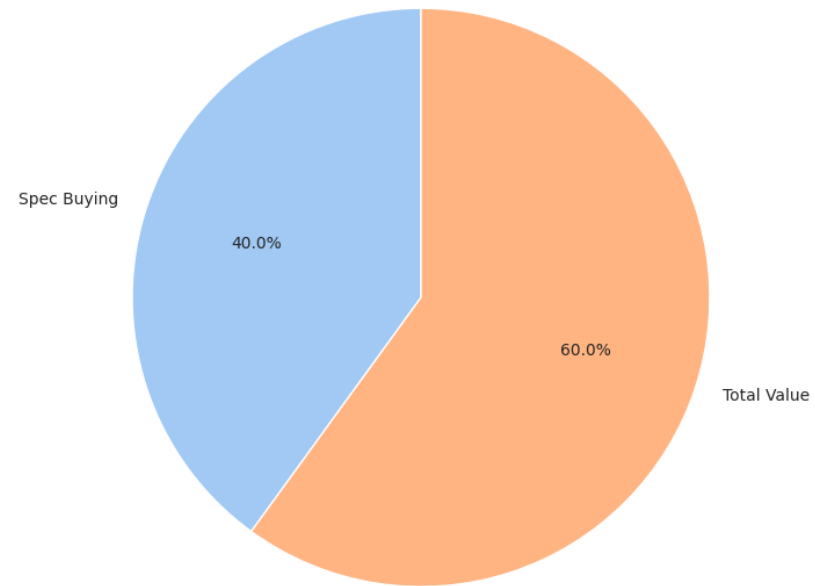


Section (1): Introduction and Variables

Spec buying:

spec_buying	Count	Percent %
Spec buying	40	40
Total value	60	60

Figure 2: Spec Buying Distribution

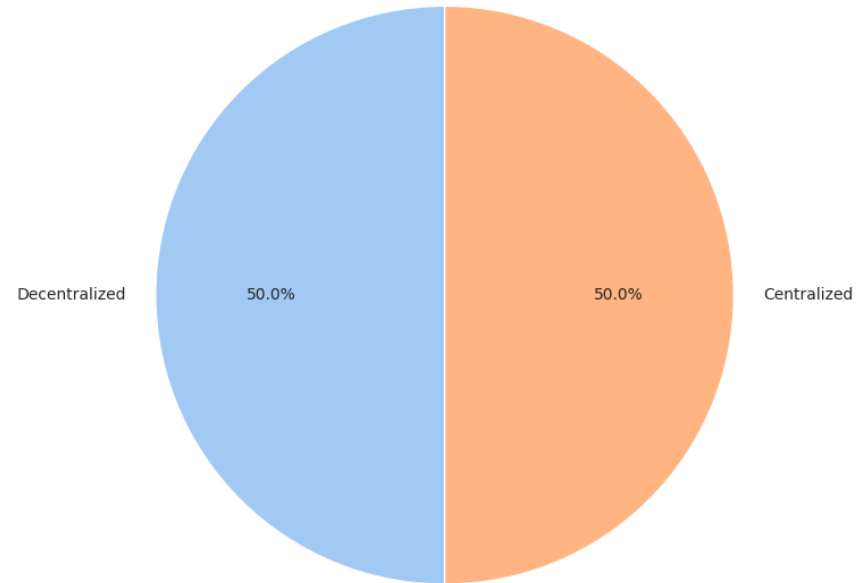


Section (1): Introduction and Variables

Procurement structure:

procurement_ structure	Count	Percent %
centralized	50	50
decentralized	50	50

Figure 3: Procurement Structure Distribution

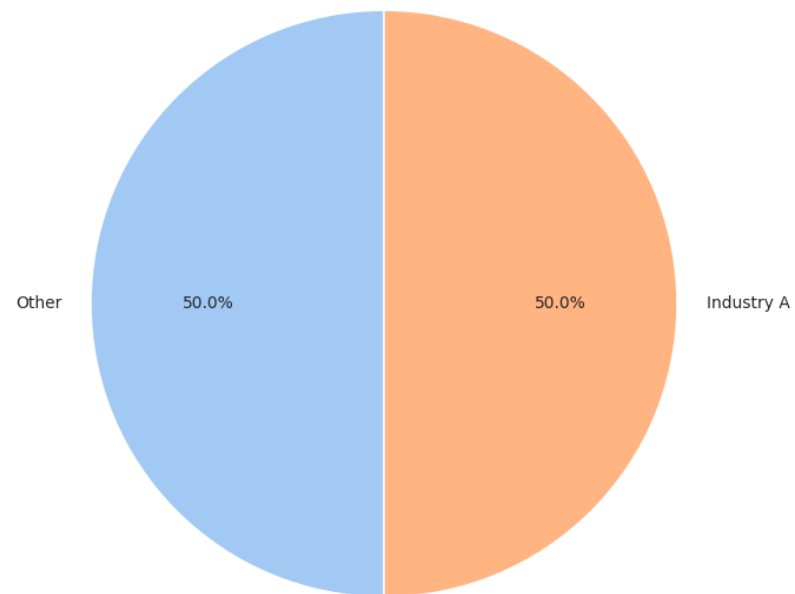


Section (1): Introduction and Variables

Industry type:

industry_type	Count	Percent %
Industry A	50	50
Other industries	50	50

Figure 4: Industry Type Distribution

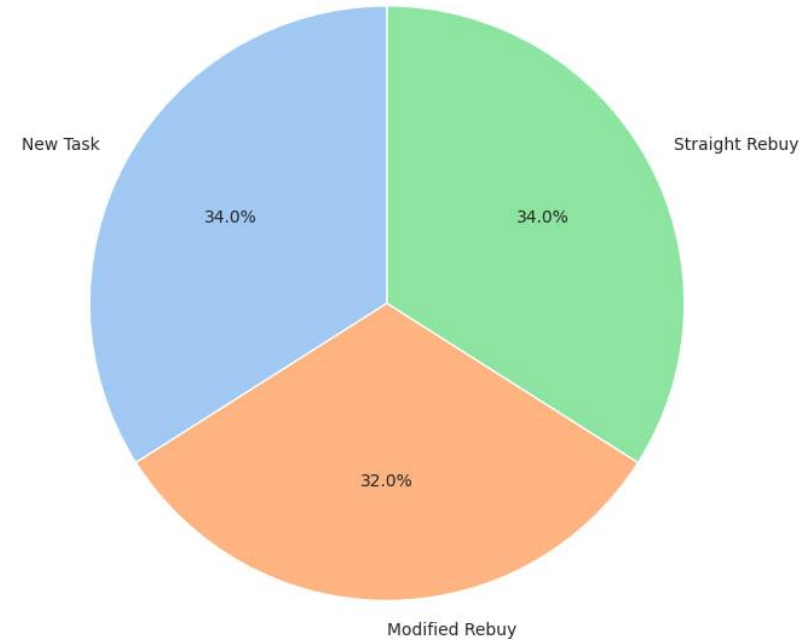


Section (1): Introduction and Variables

Buying situation:

buying_situation	Count	Percent %
New task	34	34
Modified re-buy	32	32
Straight re-buy	34	34

Figure 5: Buying Situation Distribution



Section (2): Data Visualization and Descriptive Measures

Section (2): Data Visualization and Descriptive Measures

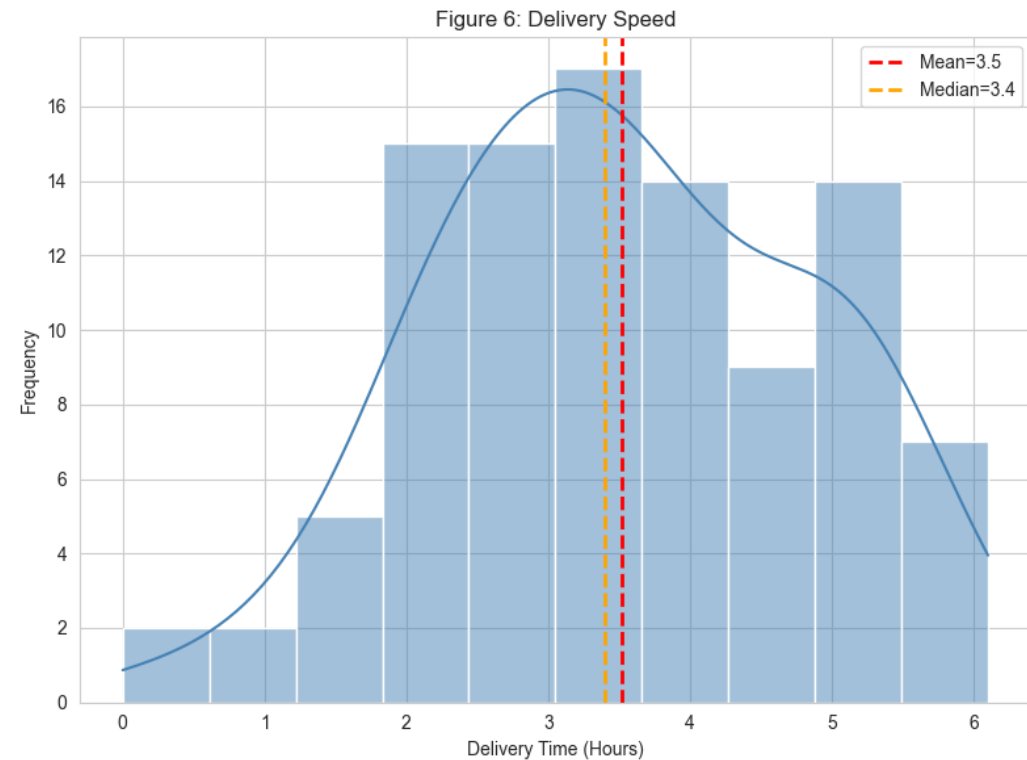
Dataset Descriptive Measures:

	count	mean	std	min	25%	50%	75%	max	range	IQR
delivery_speed	100	3.515	1.320726384	0	2.5	3.4	4.6	6.1	6.1	2.1
price_level	100	2.364	1.195658814	0.2	1.475	2.15	3.225	5.4	5.2	1.75
price_flexibility	100	7.894	1.38650203	5	6.7	8.05	9.1	10	5	2.4
manufacturer_image	100	5.248	1.131413704	2.5	4.575	5	6	8.2	5.7	1.425
service_level	100	2.916	0.751257532	0.7	2.4	3	3.425	4.6	3.9	1.025
salesforce_image	100	2.665	0.770854832	1.1	2.2	2.6	3	4.6	3.5	0.8
product_quality	100	6.971	1.585240956	3.7	5.8	7.15	8.325	10	6.3	2.525
usage_level	100	46.1	8.988769649	25	39	46.5	53.25	65	40	14.25
satisfaction	100	4.771	0.855557589	3.2	4.1	4.85	5.4	6.8	3.6	1.3

Section (2): Data Visualization and Descriptive Measures

Visualization:

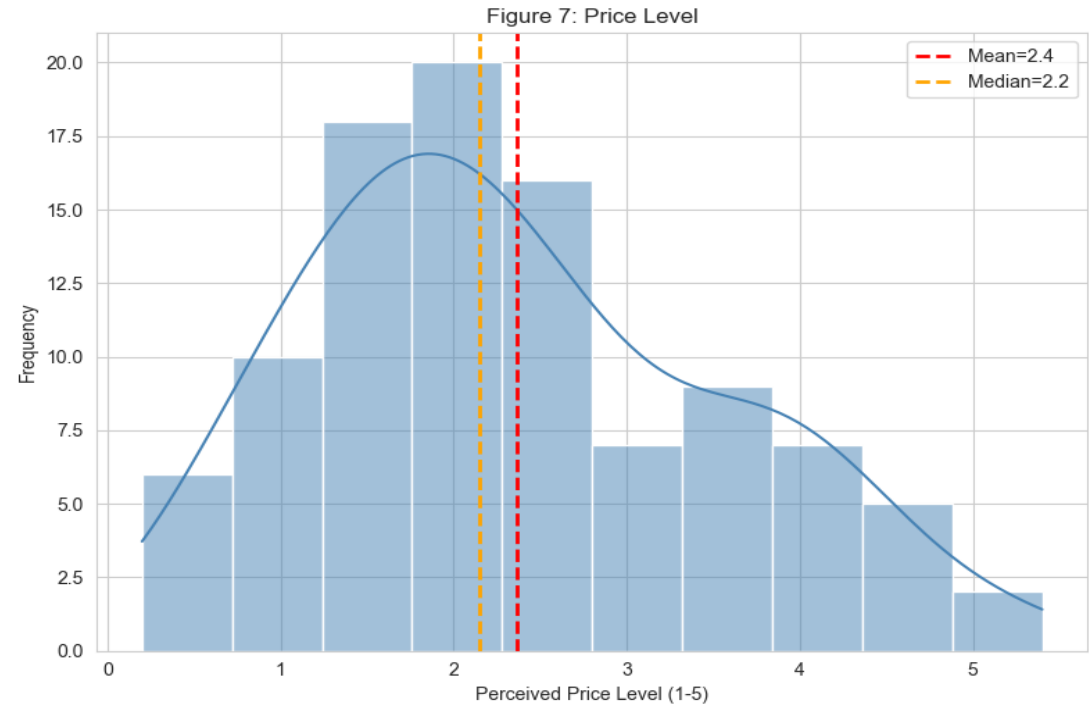
For the delivery speed the mean is 3.515 and median is 3.4 so there is slight positive skew, the standard deviation is 1.32 which indicates moderate dispersion (coefficient of variation 37.5%), the minimum is 0 hours which is logistically impossible the max is 6.1 hours which is acceptable, the IQR is 2.1 so 50% of the data lies in the area of 3.4 ± 1.05 , we can tell the customer to expect their delivery in 3.5 ± 1.3 hours.



Section (2): Data Visualization and Descriptive Measures

Visualization:

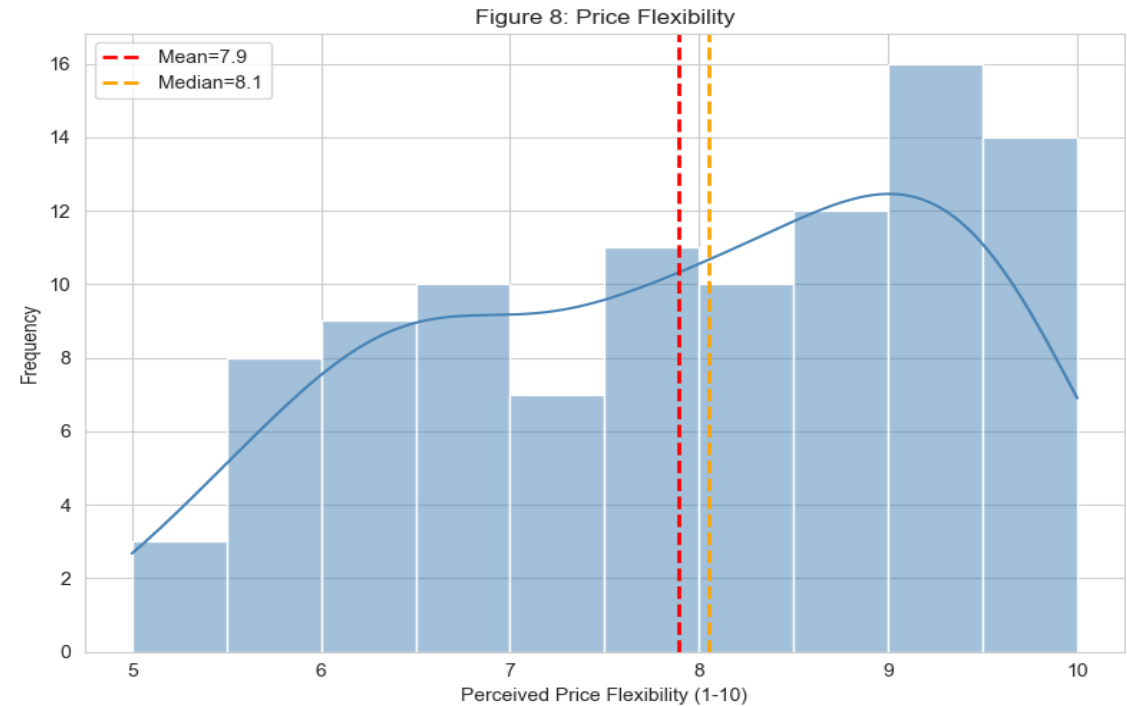
For the price level the mean is 2.364 and the median is 2.15, so there is a positive skew. The standard deviation is 1.196, which indicates high dispersion as the coefficient of variation is 50.59%. The minimum is 0.2, the max is 5.4, and the IQR is 1.75, so 50% of the data lies in the area of 2.15 ± 0.875 . We can tell management that most of the clients price evaluation fall in range 2.364 ± 1.196 which is good that the clients see that the price isn't high.



Section (2): Data Visualization and Descriptive Measures

Visualization:

For the price flexibility the mean is 7.894 and median is 8.05, so there is a slight negative skew. The standard deviation is 1.387, which indicates low dispersion as the coefficient of variation is 17.57%. The minimum is 5, the max is 10, and the IQR is 2.4, so 50% of the data lies in the area of 8.05 ± 1.2 . We can tell that the sales team is likely to negotiate and is flexible with the pricing.

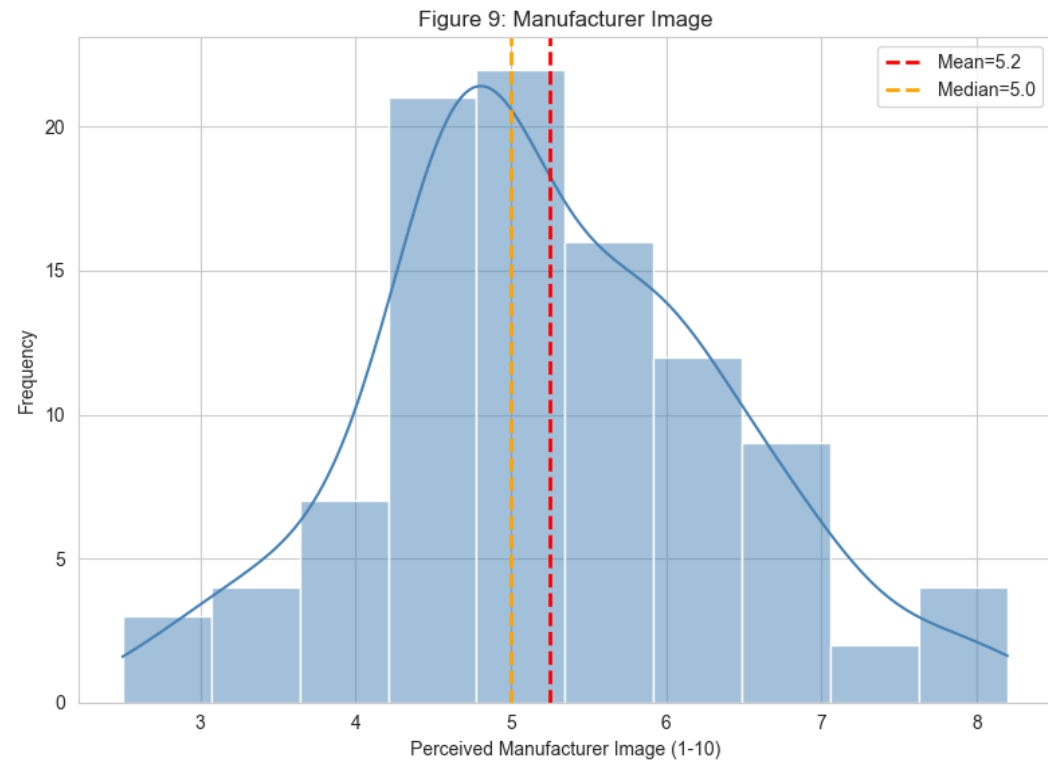


Section (2): Data Visualization and Descriptive Measures

Visualization:

For the manufacturer image the mean is 5.248 and the median is 5, so there is a slight positive skew. The standard deviation is 1.131, which indicates low dispersion as the coefficient of variation is 21.55%.

The minimum is 2.5, the max is 8.2, and the IQR is 1.425, so 50% of the data lies in the area of 5 ± 0.713 . We can tell company that its image is around 5.248 ± 1.131 on the rating scale which needs improvement.

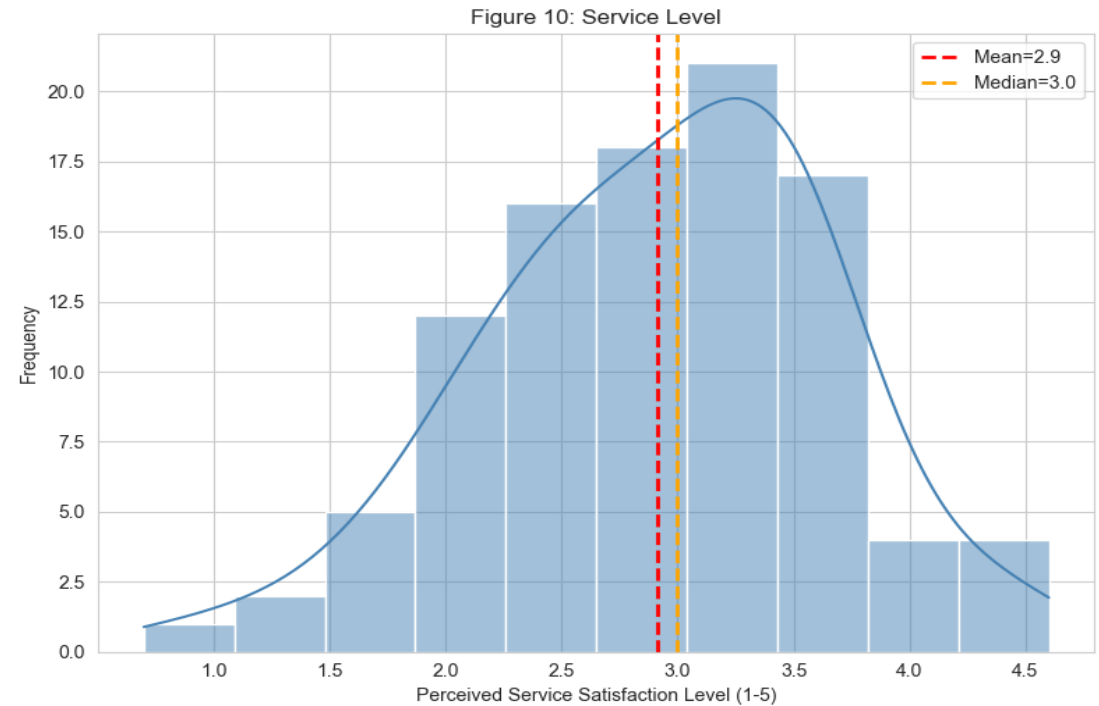


Section (2): Data Visualization and Descriptive Measures

Visualization:

For the service level the mean is 2.916 and the median is 3, so there is a slight negative skew. The standard deviation is 0.751, which indicates moderate dispersion as the coefficient of variation is 25.75%.

The minimum is 0.7 and the max is 4.6, the IQR is 1.025, so 50% of the data lies in the area of 3 ± 0.513 . We can tell that most service satisfaction scores concentrate in 2.916 ± 0.751 on the (0-5) scale.

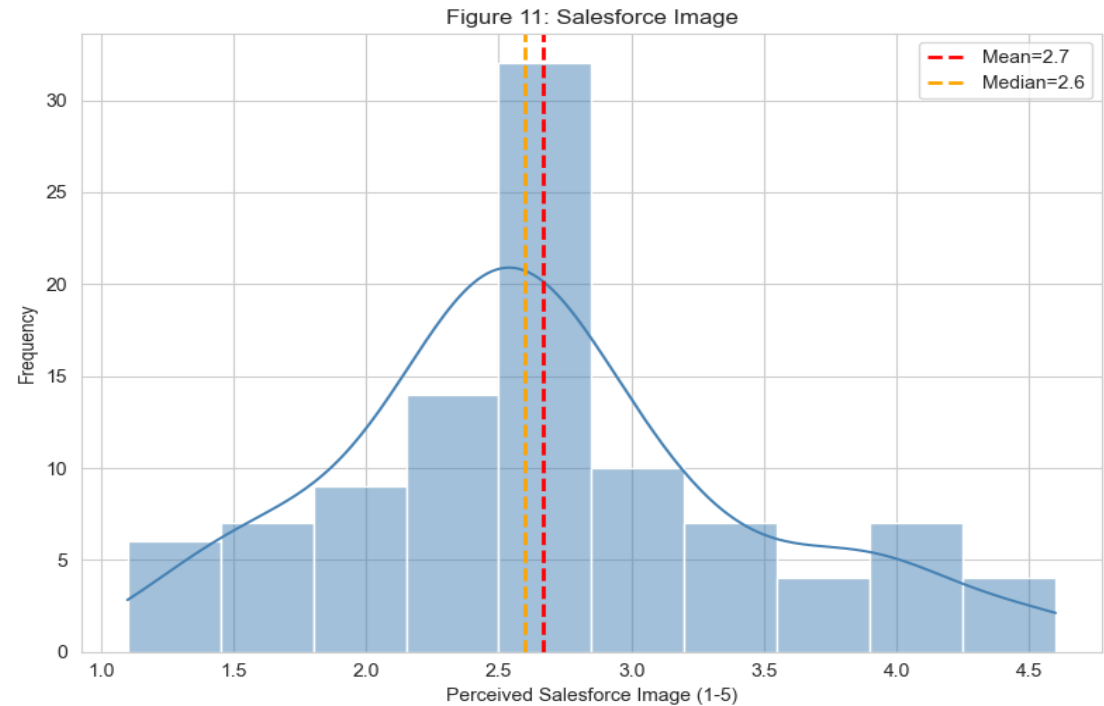


Section (2): Data Visualization and Descriptive Measures

Visualization:

For the service level the mean is 2.916 and the median is 3, so there is a slight negative skew. The standard deviation is 0.751, which indicates moderate dispersion as the coefficient of variation is 25.75%.

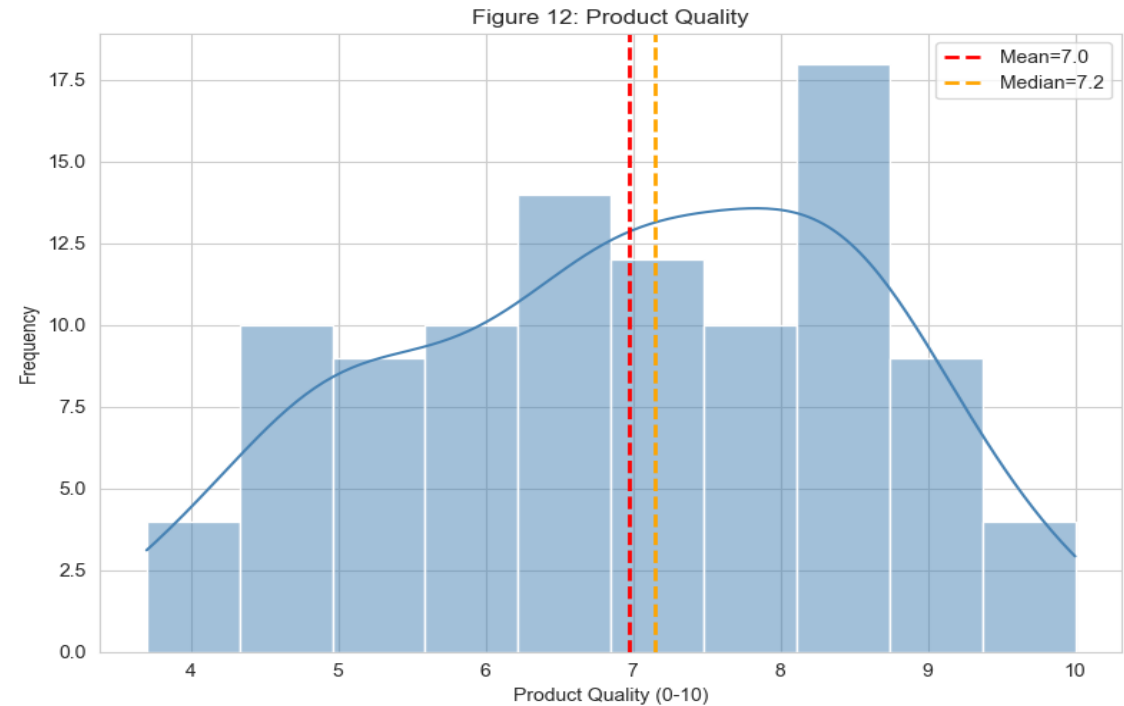
The minimum is 0.7 and the max is 4.6, the IQR is 1.025, so 50% of the data lies in the area of 3 ± 0.513 . We can tell that most service satisfaction scores concentrate in 2.916 ± 0.751 on the (0-5) scale.



Section (2): Data Visualization and Descriptive Measures

Visualization:

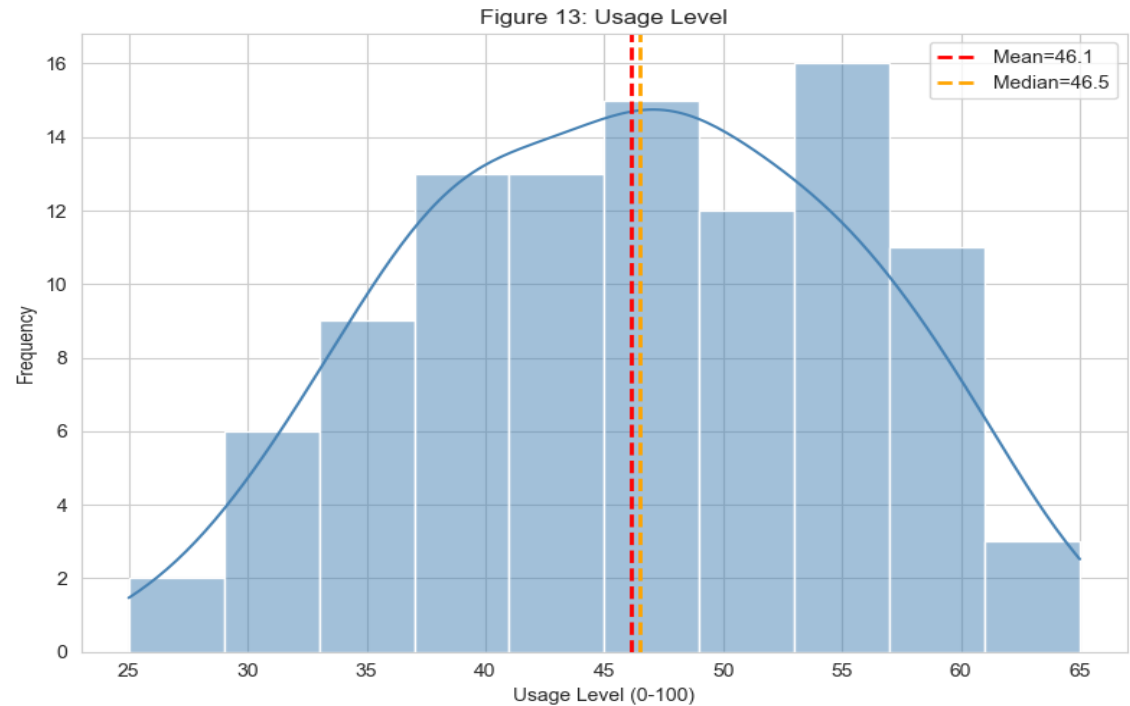
For the product quality the mean is 6.971 and median is 7.15, so there is a slight negative skew. The standard deviation is 1.585, which indicates low-to-moderate dispersion as the coefficient of variation is 22.74%. The minimum is 3.7, the max is 10, and the IQR is 2.525, so 50% of the data lies in the area of 7.15 ± 1.263 . We can inform company that quality performance clusters in 6.971 ± 1.585 on (0-10) scale which means that the clients perceive the product produced by designers and R&D team as a good product.



Section (2): Data Visualization and Descriptive Measures

Visualization:

For the usage level the mean is 46.1 and median is 46.5, so there is a nearly symmetrical, very slight negative skew. The standard deviation is 8.989, which indicates low dispersion as the coefficient of variation is 19.50%. The minimum is 25, the max is 65, and the IQR is 14.25, so 50% of the data lies in the area of 46.5 ± 7.125 . We can tell company that total purchase volume lies in $46.1\% \pm 8.989$ which leaves room for improvement.

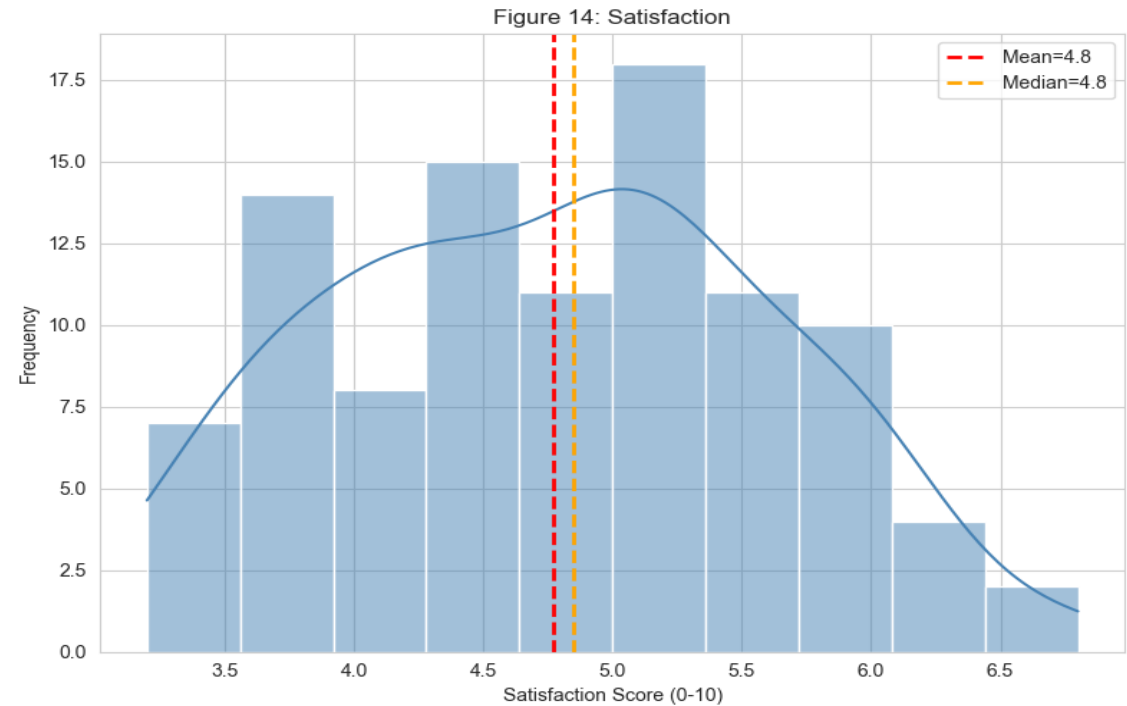


Section (2): Data Visualization and Descriptive Measures

Visualization:

For the satisfaction the mean is 4.771 and median is 4.85, so there is a slight negative skew. The standard deviation is 0.856, which indicates low dispersion as the coefficient of variation is 17.94%.

The minimum is 3.2, the max is 6.8, and the IQR is 1.3, so 50% of the data lies in the area of 4.85 ± 0.65 . We can tell company that client companies satisfaction is constricted to 4.771 ± 0.856 on a rating scale (1-10) which is really not a good indication as most of the customers have the same view based on the small coefficient of variation.



Section (3): Modeling and in-depth Analysis

ONE-WAY ANOVA

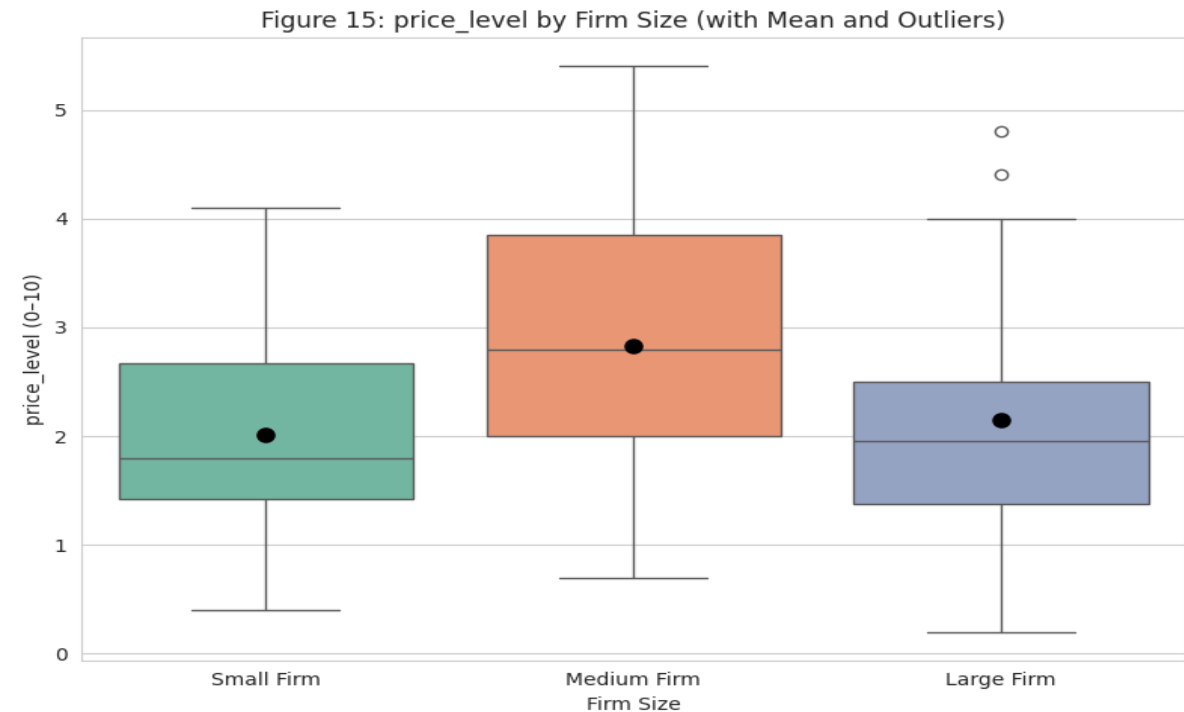
Section (3): Modeling and in-depth Analysis

A) ONE-WAY ANOVA

Does Price Level significantly affected
by Firm Size?

IV: firm_size (3 Groups)
(0=Small, 1=Medium, 2=Large)

DV: Price Level (Continuous)
Continuous (0–10)



Section (3): Modeling and in-depth Analysis

A) ONE-WAY ANOVA

Assumptions:

1. Data is Normally Distributed
2. Data is Homogeneous

Hypothesis

Ho:

The data is normally distributed in all 3 groups

Ha:

At least the data NOT is normally distributed in ONE group.

All P-Values > 0.05 in all 3 groups. So, We Accept H_0

		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	firm_size	Statistic	df	Sig.	Statistic	df	Sig.
price_level	0	.136	30	.165	.951	30	.177
	1	.093	38	.200*	.970	38	.398
	2	.153	32	.055	.954	32	.193

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Section (3): Modeling and in-depth Analysis

A) ONE-WAY ANOVA

Assumptions:

1. Data is Normally Distributed
2. Data is Homogeneous

Hypothesis

Ho:

price_level Sig2, Small = price_level Sig2, Medium = price_level Sig2, Large

Ha:

At least TWO Sig2 are NOT Equal

All P-Values > 0.05 So, We Accept Ho

Levene's Test of Equality of Error Variances^{a,b}

		Levene Statistic	df1	df2	Sig.
price_level	Based on Mean	1.581	2	97	.211
	Based on Median	1.598	2	97	.208
	Based on Median and with adjusted df	1.598	2	96.338	.208
	Based on trimmed mean	1.582	2	97	.211

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: price_level

b. Design: Intercept + firm_size

Section (3): Modeling and in-depth Analysis

A) ONE-WAY ANOVA

Model Test:

ANOVA					
price_level	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	13.554	2	6.777	5.137	.008
Within Groups	127.977	97	1.319		
Total	141.530	99			

Hypothesis

Ho:

price_level Mu, Small = price_level Mu, Medium = price_level Mu, Large

Ha:

At least TWO Mu are NOT Equal

P-Values < 0.05 So, We Reject Ho and Accept Ha

Section (3): Modeling and in-depth Analysis

A) ONE-WAY ANOVA

Post-Hoc Test:

H0: price_level Mu, Small = price_level Mu, Medium

Ha: price_level Mu, Small $\neq$ price_level Mu, Medium

P-Values = 0.012, So, We Reject H_0 and Accept H_a

H0: price_level Mu, Small = price_level Mu, Large

Ha: price_level Mu, Small $\neq$ price_level Mu, Large

P-Values = 0.881, So, We Accept H_0

H0: price_level Mu, Medium = price_level Mu, Large

Ha: price_level Mu, Medium $\neq$ price_level Mu, Large

P-Values = 0.040, So, We Reject H_0 and Accept H_a

Multiple Comparisons

Dependent Variable: price_level

	(I) firm_size	(J) firm_size	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Tukey HSD	0	1	-.822*	.2805	.012	-1.490	-.155
		2	-.140	.2919	.881	-.835	.555
	1	0	.822*	.2805	.012	.155	1.490
		2	.682*	.2756	.040	.026	1.338
	2	0	.140	.2919	.881	-.555	.835
		1	-.682*	.2756	.040	-1.338	-.026

Based on observed means.
The error term is Mean Square(Error) = 1.319.
*. The mean difference is significant at the .05 level.

Section (3):
Modeling and in-depth Analysis

ONE-WAY MANOVA

Section (3): Modeling and in-depth Analysis

B) ONE-WAY MANOVA

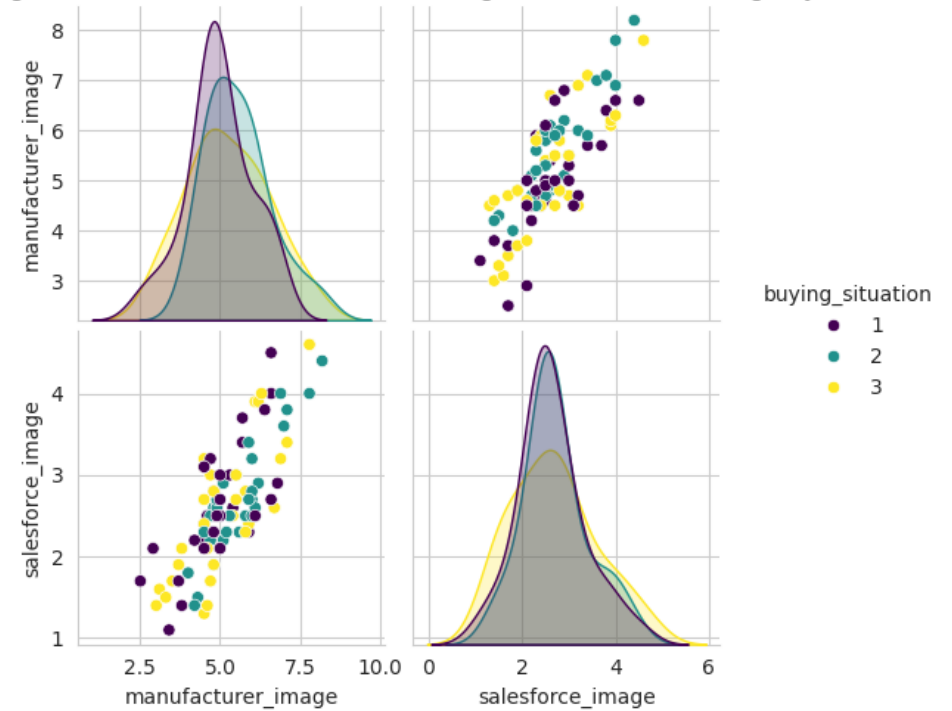
Do Salesforce Image and Manufacturer Image combined significantly affected by Buying Situation?

IV: buying_situation(3 Groups)
(0=New task, 1=Modified re-buy, 2=Straight re-buy)

DVs: (Continuous)

1. salesforce_image
Continuous (0–5)
2. manufacturer_image
Continuous (0–10)

Figure 17: Pair Plot of manufacturer image and salesforce image by Firm Size



Section (3): Modeling and in-depth Analysis

B) ONE-WAY MANOVA

Assumptions:

1. Data is Normally Distributed
2. Data is Homogeneous
3. Covariance is Homogeneous

Hypothesis

Ho:

The data for dependent variables are normally distributed in all 3 groups

Ha:

The data for dependent variables are NOT is normally distributed in at least in ONE group.

All P-Values > 0.05. So, we Accept Ho

		Tests of Normality					
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	buying_situation	Statistic	df	Sig.	Statistic	df	Sig.
salesforce_image	1	.129	34	.167	.962	34	.286
	2	.163	32	.030	.945	32	.104
	3	.090	34	.200 [*]	.956	34	.190
manufacturer_image	1	.160	34	.027	.955	34	.169
	2	.117	32	.200 [*]	.941	32	.077
	3	.134	34	.125	.969	34	.445

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Section (3): Modeling and in-depth Analysis

B) ONE-WAY MANOVA

Assumptions:

1. Data is Normally Distributed
2. Data is Homogeneous
3. Covariance is Homogeneous

Hypothesis

Ho:

Combined DVs Sig2, New = Combined DVs Sig2, Modified = Combined DVs Sig2, Straight

Ha:

At least Combined DVs Sig2 of TWO group are NOT Equal.

All P-Values > 0.05. So, We Accept Ho

Levene's Test of Equality of Error Variances^a

		Levene Statistic	df1	df2	Sig.
manufacturer_image	Based on Mean	2.030	2	97	.137
	Based on Median	1.999	2	97	.141
	Based on Median and with adjusted df	1.999	2	96.014	.141
	Based on trimmed mean	2.005	2	97	.140
salesforce_image	Based on Mean	1.735	2	97	.182
	Based on Median	1.779	2	97	.174
	Based on Median and with adjusted df	1.779	2	96.058	.174
	Based on trimmed mean	1.740	2	97	.181

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Design: Intercept + buying_situation

Section (3): Modeling and in-depth Analysis

B) ONE-WAY MANOVA

Assumptions:

1. Data is Normally Distributed
2. Data is Homogeneous
3. Covariance is Homogeneous

Hypothesis

Ho:

DVs Covariance, New = DVs Covariance, Modified = DVs Covariance, Straight

Ha:

At least DVs Covariance of TWO group are NOT Equal.

P-Values > 0.05. So, We Accept Ho

**Box's Test of
Equality of
Covariance
Matrices^a**

Box's M	12.891
F	2.084
df1	6
df2	229276.444
Sig.	.052

Tests the null hypothesis that the observed covariance matrices of the dependent variables are equal across groups.

a. Design:
Intercept +
buying_situation

Section (3): Modeling and in-depth Analysis

B) ONE-WAY MANOVA

Model Test:

Multivariate Tests ^a						
Effect		Value	F	Hypothesis df	Error df	Sig.
Intercept	Pillai's Trace	.959	1112.020 ^b	2.000	96.000	<.001
	Wilks' Lambda	.041	1112.020 ^b	2.000	96.000	<.001
	Hotelling's Trace	23.167	1112.020 ^b	2.000	96.000	<.001
	Roy's Largest Root	23.167	1112.020 ^b	2.000	96.000	<.001
buying_situation	Pillai's Trace	.087	2.198	4.000	194.000	.071
	Wilks' Lambda	.913	2.226 ^b	4.000	192.000	.068
	Hotelling's Trace	.095	2.254	4.000	190.000	.065
	Roy's Largest Root	.095	4.600 ^c	2.000	97.000	.012

a. Design: Intercept + buying_situation

b. Exact statistic

c. The statistic is an upper bound on F that yields a lower bound on the significance level.

Hypothesis

Ho:

Combined DVs Mu, New = Combined DVs Mu, Modified = Combined DVs Mu, Straight

Ha:

At least TWO Combined DVs Mu are NOT Equal

P-Values for BOTH Wilks' Lambda and Pillai's > 0.05 So, We Accept Ho

Section (3): Modeling and in-depth Analysis

B) ONE-WAY MANOVA

As MANOVA test Passed
NO Need for Post-Hoc Test

buying_situation

Tukey HSD

Multiple Comparisons

Dependent Variable	(I) buying_situation	(J) buying_situation	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
manufacturer_image	1	2	-.607	.2747	.075	-1.261	.047
		3	-.279	.2705	.558	-.923	.364
	2	1	.607	.2747	.075	-.047	1.261
		3	.327	.2747	.461	-.326	.981
	3	1	.279	.2705	.558	-.364	.923
		2	-.327	.2747	.461	-.981	.326
salesforce_image	1	2	-.098	.1915	.866	-.554	.358
		3	-.056	.1886	.953	-.505	.393
	2	1	.098	.1915	.866	-.358	.554
		3	.042	.1915	.974	-.414	.498
	3	1	.056	.1886	.953	-.393	.505
		2	-.042	.1915	.974	-.498	.414

Based on observed means.

The error term is Mean Square(Error) = .605.